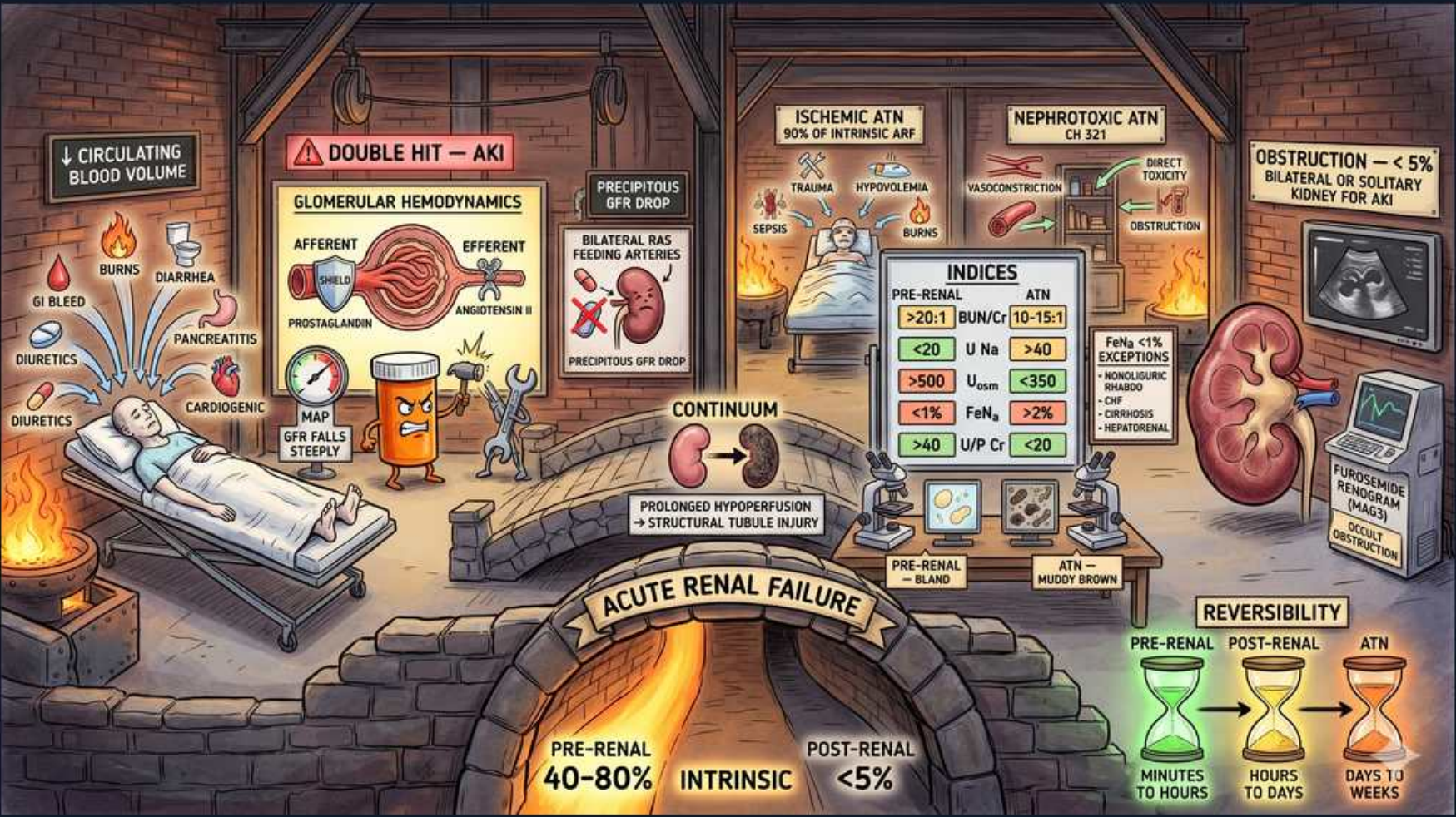


Renal Approach — Azotemia / Uremia Overview (Pre/Intra/Post-renal & BUN:Cr)



1. Acute Renal Failure arch

WHAT YOU SEE

Stone bridge labeled ACUTE RENAL FAILURE spanning the scene

WHAT IT ENCODES

AKI is an abrupt (hours-days) fall in GFR defined by KDIGO as a rise in creatinine ≥ 0.3 mg/dL in 48h, a $\geq 1.5\times$ rise over baseline within 7 days, or urine output <0.5 mL/kg/h for 6h. Every AKI is triaged into three buckets: pre-renal (~40-80% of cases), intrinsic, and post-renal (<5%). 'Azotemia' just means elevated nitrogenous waste (BUN/Cr); 'uremia' is the symptomatic syndrome (encephalopathy, pericarditis, platelet dysfunction, asterixis). Mnemonic: think Pre (perfusion), Intra (parenchyma), Post (plumbing).

BOARD TRAP

Tempting wrong answer: 'azotemia = intrinsic kidney disease, biopsy the kidney.' Azotemia alone does NOT localize the lesion — the discriminator is to first place the patient into pre/intra/post-renal using volume status, urine indices, and a bladder scan/renal ultrasound before ever considering biopsy.

2. Decreased circulating volume

WHAT YOU SEE

Top-left tap with drop and list: burns, GI bleed, diarrhea, diuretics, cardiogenic, pancreatitis

WHAT IT ENCODES

Pre-renal azotemia is hypoperfusion from either true volume loss (hemorrhage, GI losses, burns, over-diuresis, third-spacing in pancreatitis) or low EFFECTIVE circulating volume despite total-body overload (CHF/cardiogenic shock, cirrhosis, nephrotic syndrome). The kidney senses low perfusion and avidly reabsorbs Na and water, concentrating the urine. It is the single most common cause of AKI and is rapidly reversible if perfusion is restored with fluids (or treating the heart/liver). Mnemonic — 'the kidney is crying out for blood, not broken.'

BOARD TRAP

Tempting wrong answer: 'edematous CHF/cirrhosis patient can't be pre-renal because they're fluid-overloaded.' The discriminator is EFFECTIVE arterial volume — in CHF and cirrhosis the kidney is underperfused despite total-body fluid excess, so they show pre-renal indices (FENa $<1\%$, BUN:Cr >20).

3. Glomerular hemodynamics

WHAT YOU SEE

Steak (afferent) and steak with arrow (efferent) under GLOMERULAR HEMODYNAMICS

WHAT IT ENCODES

GFR depends on the pressure gradient across the glomerulus set by afferent vs efferent arteriolar tone. Prostaglandins dilate the AFFERENT arteriole (NSAIDs block this $\rightarrow$ afferent constriction $\rightarrow$ $\downarrow$ GFR); angiotensin II constricts the EFFERENT arteriole to hold filtration pressure (ACEi/ARB block this $\rightarrow$ efferent dilation $\rightarrow$ $\downarrow$ GFR). In a normally perfused kidney these drugs are tolerated, but during hypoperfusion they remove the compensatory mechanism and unmask pre-renal AKI. Mnemonic: 'A for Afferent, NSAIDs Attack the Afferent; E for Efferent, ACEi relax the Efferent.'

BOARD TRAP

Tempting wrong answer: swapping the arterioles — 'ACEi constrict the afferent.' Discriminator: NSAIDs constrict the AFFERENT (block vasodilatory prostaglandins), ACEi/ARB dilate the EFFERENT (block AngII). The triple-whammy of NSAID + ACEi/ARB + diuretic in a volume-depleted patient is the classic exam setup for hemodynamic AKI.

4. MAP / GFR falls steeply

WHAT YOU SEE

Angry pill bottle + MAP gauge: GFR FALLS STEEPLY

WHAT IT ENCODES

Renal autoregulation keeps GFR roughly constant across a MAP of ~80-180 mmHg via myogenic reflex and tubuloglomerular feedback. Below the lower autoregulatory limit (MAP ~70-80, or higher in chronic HTN where the curve is shifted right), GFR falls steeply and linearly with pressure. This is why hypotensive/septic patients and chronically hypertensive patients drop GFR sooner. Mnemonic — once you fall off the autoregulation plateau, the cliff is steep.

BOARD TRAP

Tempting wrong answer: 'a chronically hypertensive patient is protected from hypoperfusion AKI because their pressures run high.' Opposite — their autoregulatory curve is shifted right, so a 'normal' MAP of 75 that a young patient tolerates can already be below their threshold and precipitate ischemic injury.

5. Double hit / precipitous drop

WHAT YOU SEE

Warning triangle DOUBLE HIT $\rightarrow$ AKI and PRECIPITOUS GFR DROP with bilateral RAS arteries

WHAT IT ENCODES

Bilateral renal artery stenosis (or stenosis of a solitary functioning kidney) makes the glomerulus dependent on efferent (AngII) constriction to maintain filtration pressure against a fixed low inflow. Add an ACEi/ARB and the efferent dilates, collapsing the gradient $\rightarrow$ precipitous, often $>30\%$, rise in creatinine within days. Causes of bilateral RAS: atherosclerosis (older, vascular) and fibromuscular dysplasia (young women, 'string of beads'). Mnemonic — 'stenosis + ACEi = the floor drops out.'

BOARD TRAP

Tempting wrong answer: 'any ACEi-induced creatinine bump means bilateral RAS — stop the drug.' Discriminator: a mild $\leq 30\%$ creatinine rise after starting an ACEi is expected, reflects efferent dilation, and you continue the drug; only a sharp $>30\%$ rise (or hyperkalemia) suggests bilateral RAS. Also, UNILATERAL RAS with a healthy contralateral kidney does NOT cause this — the good kidney compensates.

6. Continuum to ATN

WHAT YOU SEE

CONTINUUM: prolonged hypoperfusion $\rightarrow$ structural tubule injury

WHAT IT ENCODES

Pre-renal azotemia and ischemic ATN are two ends of one continuum, not separate diseases. Brief hypoperfusion is purely functional and reverses with fluids; if ischemia is prolonged the energy-starved tubular cells die, slough, and form casts — now it is structural ATN that will NOT reverse with fluids alone and takes days-weeks to recover. The proximal tubule (S3 segment, outer medulla) and thick ascending limb are most vulnerable because they have the highest metabolic demand. Mnemonic — 'pre-renal ignored long enough becomes ATN.'

BOARD TRAP

Tempting wrong answer: 'give fluids and the creatinine will correct in hours' applied to a patient who has already crossed into ATN. Discriminator: if FENa is now $>2\%$, urine has muddy-brown casts, and creatinine keeps climbing despite euolemia, the tubules are dead — fluids won't fix it; over-resuscitating risks pulmonary edema while you wait days-weeks for tubular regeneration.

7. Ischemic ATN

WHAT YOU SEE

ISCHEMIC ATN 90% of intrinsic AKI — trauma, sepsis, hypovolemia

WHAT IT ENCODES

Ischemic ATN is the most common cause of intrinsic AKI (~90% of intrinsic cases) and follows shock, sepsis, major surgery/trauma, or prolonged hypovolemia. Tubular cells undergo ATP depletion → loss of polarity, cast formation, and back-leak of filtrate. The hallmark sediment is muddy-brown granular and renal tubular epithelial cell casts; clinically it follows initiation → maintenance (oliguric, hyperkalemia/acidosis peak) → recovery (often a polyuric diuretic phase). Management is supportive — there is no proven pharmacologic reversal; avoid further nephrotoxins.

BOARD TRAP

Tempting wrong answer: 'AKI after sepsis with active sediment must be glomerulonephritis — order ANCA/complement and biopsy.' Discriminator: most intrinsic AKI is ATN, not GN. Muddy-brown granular casts = ATN; you need RBC casts and dysmorphic RBCs for GN. The clinical context (shock/sepsis/contrast/nephrotoxin) plus FENa >2% nails ATN.

8. Nephrotoxic ATN

WHAT YOU SEE

NEPHROTOXIC ATN — direct toxicity, vasoconstriction, obstruction

WHAT IT ENCODES

Nephrotoxic ATN comes from direct tubular toxins, intrarenal vasoconstriction, or intratubular obstruction. Exogenous: aminoglycosides (nonoliguric ATN ~5-7 days in, dose by levels), amphotericin B, cisplatin, IV contrast, tenofovir. Endogenous pigments: myoglobin (rhabdomyolysis — CK often >5,000-10,000, dipstick + for blood with no RBCs) and hemoglobin; crystals: tumor lysis (uric acid), ethylene glycol (calcium oxalate), acyclovir, methotrexate. Prevention: volume expansion and avoiding stacking nephrotoxins. Mnemonic — 'AAA + Contrast + Pigments' (Aminoglycosides, Amphotericin, cisplAtin).

BOARD TRAP

Tempting wrong answer: calling pigment-induced (rhabdo/hemolysis) AKI 'pre-renal' because FENa can be <1%. Discriminator: heme-pigment ATN is a classic FENa <1% EXCEPTION — look for CK elevation, tea-colored urine, dipstick-positive-blood-with-no-RBCs, and hyperkalemia/hyperphosphatemia out of proportion. Treat with aggressive IV fluids, not just observation.

9. BUN:Cr ratio

WHAT YOU SEE

INDICES board: PRE-RENAL BUN/Cr >20:1 vs ATN 10-15:1

WHAT IT ENCODES

In pre-renal states the kidney avidly reabsorbs Na and water in the proximal tubule, and urea passively follows water, so BUN rises out of proportion to creatinine → BUN:Cr >20:1. In ATN the tubules can't reabsorb, so urea is lost with everything else and the ratio normalizes to ~10-15:1. Normal ratio is ~10-15:1. Mnemonic — 'pre-renal kidney hoards water and urea comes along for the ride.'

BOARD TRAP

Tempting wrong answer: 'BUN:Cr >20 always means pre-renal, give fluids.' Discriminator: a high BUN:Cr also occurs WITHOUT hypoperfusion in GI bleeding (blood = protein load), high-protein feeds, corticosteroids/catabolic states, and tetracyclines — all raise BUN independent of GFR. Conversely the ratio is falsely LOW in malnutrition, liver disease, and SIADH. Always pair it with FENa and urine osmolality.

10. FENa

WHAT YOU SEE

Indices: FENa <1% pre-renal vs >2% ATN, with exceptions noted

WHAT IT ENCODES

FENa = $(\text{UNa} \times \text{PCr}) / (\text{PNa} \times \text{UCr}) \times 100$ measures the fraction of filtered sodium excreted. Pre-renal kidneys are perfusion-starved and reabsorb Na avidly → FENa <1% (UNa <20). ATN tubules are injured and waste Na → FENa >2% (UNa >40). FENa is only valid in OLIGURIC AKI and is invalidated by diuretics — in a diuretic-treated patient use FEurea instead (FEurea <35% = pre-renal). Mnemonic — 'low FENa, kidney is starving for volume.'

BOARD TRAP

Tempting wrong answer: 'FENa <1% always equals pre-renal.' Discriminator — memorize the FENa <1% EXCEPTIONS (intrinsic causes that still retain Na): contrast nephropathy, early/acute obstruction, acute glomerulonephritis, hepatorenal syndrome, and pigment (rhabdo/hemoglobin) AKI. Conversely FENa can be falsely HIGH in a pre-renal patient on diuretics or with CKD — switch to FEurea.

11. Urine osmolality / Na

WHAT YOU SEE

Indices: U Osm >500 pre-renal vs <350 ATN; U Na <20 vs >40

WHAT IT ENCODES

A well-perfused but volume-starved (pre-renal) kidney maximally concentrates urine: Uosm >500 mOsm/kg and UNa <20 mEq/L because ADH-driven water reabsorption and Na avidity are intact. ATN destroys the concentrating gradient and tubular Na handling, so urine becomes isosthenuric (Uosm ~300-350, similar to plasma) with UNa >40. The concentrating ability is the functional readout of an intact tubule. Mnemonic — 'pre-renal pees concentrated salt-poor urine; ATN pees dilute salty urine.'

BOARD TRAP

Tempting wrong answer: 'a Uosm of 320 with polyuria points to diabetes insipidus.' Discriminator: in the AKI context an isosthenuric Uosm (~300) means the tubule has lost concentrating power = ATN, not DI. DI presents as polyuria in a non-AKI patient with hypernatremia and a Uosm that stays low after water deprivation.

12. Muddy brown casts

WHAT YOU SEE

ATN — MUDDY BROWN granular casts on the indices board

WHAT IT ENCODES

Muddy-brown granular casts and renal tubular epithelial cell casts are the pathognomonic sediment of ATN — they are sloughed, degenerating tubular cells packed into Tamm-Horsfall protein. Their presence in an AKI workup confirms intrinsic tubular injury and argues against a purely pre-renal or post-renal process. The more casts and RTE cells, the more severe the ATN. Mnemonic — 'mud in the urine = dead tubule.'

BOARD TRAP

Tempting wrong answer: confusing casts on the boards. Discriminator — muddy-brown granular casts = ATN; RBC casts/dysmorphic RBCs = glomerulonephritis; WBC casts = pyelonephritis or AIN; broad waxy casts = chronic CKD; fatty/oval-fat-body 'maltese cross' casts = nephrotic-range proteinuria. RBC casts are NEVER from ATN.

13. Post-renal obstruction

WHAT YOU SEE

OBSTRUCTION — <5%, bilateral or solitary kidney for AKI; stone/prostate causes

WHAT IT ENCODES

Post-renal AKI (<5% of cases) requires obstruction of BOTH kidneys or of a solitary functioning kidney to raise creatinine, since one healthy kidney can compensate. Common causes: BPH/prostate cancer (bladder outlet — most common in older men), bilateral ureteral stones, retroperitoneal fibrosis/malignancy, neurogenic bladder, and anticholinergic-induced retention. First test is a post-void residual / bladder scan and renal ultrasound looking for hydronephrosis; relief is bladder catheter or ureteral stent/nephrostomy. Mnemonic — 'plug both drains, or the only drain, to back up the creatinine.'

BOARD TRAP

Tempting wrong answer: 'unilateral ureteral stone in a patient with two kidneys explains the AKI.' Discriminator: unilateral obstruction with a normal contralateral kidney keeps creatinine NORMAL (the other kidney compensates) — look instead for a bladder-outlet cause (BPH, retention) or bilateral process. Also, early obstruction can show FENa <1% (an exception) and ultrasound may miss hydronephrosis in the first 24-48h or with retroperitoneal encasement.

14. Reversibility hourglasses

WHAT YOU SEE

REVERSIBILITY: pre-renal minutes-hours, post-renal hours-days, ATN days-weeks

WHAT IT ENCODES

Recovery timeline tracks how much structural damage occurred: pre-renal is functional and corrects in minutes-hours once perfusion is restored; post-renal recovers in hours-days after the obstruction is relieved (watch for post-obstructive diuresis and persistent tubular dysfunction if prolonged); ATN takes days-weeks because the tubular epithelium must regenerate, often with an intervening oliguric phase and a recovery polyuric phase. Mnemonic — 'fix the pipe fast, fix the plug medium, regrow the tubule slow.'

BOARD TRAP

Tempting wrong answer: 'relieving the obstruction should immediately normalize creatinine, so if it doesn't the diagnosis was wrong.' Discriminator: after relieving prolonged obstruction, expect a post-obstructive diuresis and gradual (hours-days) creatinine improvement, plus possible lasting concentrating/acidification defects — slow recovery does not invalidate the post-renal diagnosis. Monitor for volume depletion and electrolyte losses during the diuretic phase.